

PSYCHOLOGY 101 · ANSWER KEY - PROFESSOR COPY

Homework 2: Bonding

Grading guide. Total 21 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Three experiments

1. Fill in the table. (1 point for what was measured, 1 point for what it showed, per row.) (6 pts)

Answer: Harlow: hours spent on each pretend mother and where the monkeys ran when scared -> comfort matters more than food. Ainsworth: what the baby does when the mother returns -> a reliably responsive caregiver builds a secure rope (about 6 in 10). Tronick: what the baby does when the game stops -> babies expect serve-and-return and work to restart it.

Part B — Serve and return, tallied

2. Watch a grown-up and a baby or toddler for two minutes (a real one, or a video — 'still face experiment' or any parent-and-baby clip works; a pet and its person is fine too). Make a tally mark every time the little one 'serves' (a look, sound, reach, smile) and another every time the grown-up 'returns' it. Write your two totals, then one sentence about what you noticed. (4 pts)

Answer: Two tally totals (2 pts — any honest numbers; typically dozens per two minutes). A sentence of observation (2 pts): e.g., returns came fast; the baby served again right after a return; when a return was missed the baby tried harder or looked away.

Part C — Secure or not?

3. Baby R cries when her mother leaves the strange-situation room, runs to her when she comes back, settles in about a minute, then goes back to the blocks. Ainsworth would call this... (1 pt)

Answer: A. secure — the rope is strong and the baby knows it. - The tell is the reunion: comforted quickly, then back to exploring. Crying at the leaving is normal.

4. Baby S does not look up when his mother leaves, and does not go to her when she returns; he keeps playing with his back to her. Ainsworth found babies like S usually had caregivers who did what? What is the ONE word that matters most in her answer? (2 pts)

Answer: Their signals were often not responded to, so they learned not to expect a return (1 pt). The word: RELIABLY — a secure rope comes from a caregiver who responds reliably, not perfectly (1 pt).

Part D — Keeping the rope in a NICU

5. List three things a NICU can do to keep the rope between a preemie and her parents strong — and for each, say what a scientist could **MEASURE** to prove it works. (3 pts)

Answer: Any three of: kangaroo care / skin-to-skin (measure heart rate, temperature, breathing pauses, survival); unlimited visiting hours (measure how many hours parents are present, parent stress scores); parents doing the care — feeds, changes (measure weight gain, parent anxiety); a bed or chair to sleep in, phone by the isolette, reading to the baby (measure hearing-brain development, time to full feeds). 1 pt per item with a measurement.

Part E — Flashbacks

6. From NICU Tutor: name one thing kangaroo care does for a baby's **BODY** (not its feelings). (1 pt)

Answer: Keeps it warm; steadies heart rate and breathing; helps feeding. Any one.

7. From last week's Psychology: what tool did Robert Fantz invent for finding out what babies prefer? (1 pt)

Answer: Looking time — timing how long a baby stares at one picture versus another.

Part F — Think like a doctor

8. A mother says to you on rounds: “I feel like a visitor, not a mom. She's in a box with wires, and I'm not allowed to do anything.” Her baby is on CPAP in an isolette, stable. Using today's class, what would you say, and what would you change about her day? (3 pts)

Answer: Say: the rope is real and it's stretched by the machines, not by her; the science says her chest is a treatment (1 pt). Change: daily kangaroo care on CPAP (stable babies can do it), let her do the hands-on care — diaper changes, temperature checks, tube feeds, reading aloud — so she's on the team, not visiting; give her a chair that reclines, unlimited hours, a phone number for the night (1 pt). Why: measured outcomes — steadier vitals, better weight gain, less parent anxiety, and in the 2021 study, fewer deaths (1 pt).

HOME LAB (optional, not graded): The still face, grown-up version

You need: a willing grown-up with a good poker face and a timer (NOT a baby — that's a scientists-only lab)

1. Talk with the grown-up normally for 30 seconds while they respond as usual.
2. On your signal, they go still-faced for 30 seconds: no expression, no nodding, no words. You keep talking.
3. Notice what you do. Louder? A joke? Waving? Giving up? Notice how it feels.
4. Swap roles: you go still-faced while they talk. Watch what THEY do.
5. Write three things each of you did to restart the game.

What you should see: Both of you 'serve harder' — talk louder, make faces, wave, laugh, ask 'are you okay?' — then give up or look away. It feels surprisingly bad, fast. That's the still-face effect: even grown-ups expect serve-and-return, and babies are built to fight for it.